Name: Lourdes Amaya Sibrián		
School: Academica Británica Cuscatleca		
Group: 5th Trus…		**Sex:** Female
Date of first test: 14/11/2024	**Level:** 9B	**Age:** 10:04
Date of second test: 13/10/2025	**Level:** 10B	**Age:** 11:03

Scores

No. attempted (/58)	SAS	SAS (with 90% confidence bands) 60 70 80 90 100 110 120 130 140	Overall ST	PR	GR (/24)	End of KS2 indicator	Progress Category
58	117		7	87	3	111	Higher

Previous & Current Scores

Date	Age at test (yrs:mths)	Level / Form	No. attempted	SAS	SAS (with 90% confidence bands) 60 70 80 90 100 110 120 130 140	ST	PR	End of KS2 indicator
08/05/2025	10:10	10A	56 / 58	110		6	74	108
13/10/2025	11:03	10B	58 / 58	117		7	87	111

Analysis of Curriculum content categories

Curriculum content category	Number of questions	Student % correct	Standardisation % correct	Student / standardisation difference
Number	34	80%	54%	26%
Measurement	10	91%	36%	55%
Geometry	6	71%	51%	20%
Statistics	8	75%	66%	9%

Analysis of Process categories

Process category	Number of questions	Student % correct	Standardisation % correct	Student / standardisation difference
Fluency in facts and procedures	9	89%	63%	26%
Fluency in conceptual understanding	22	73%	52%	21%
Problem solving	5	83%	43%	40%
Mathematical reasoning	22	84%	50%	34%

Implications for teaching and learning

- By comparing scores from a previous administration of *PTM* it is possible to categorise progress as expected, higher or lower than expected or much higher or much lower than expected.
  - Lourdes took *PTM9B* in November 2024 and from then until now has made higher than expected progress in maths.

- Reviewing the *Analysis of Curriculum content categories* will help to identify where there are specific strengths and weaknesses and to plan next steps.
- Where performance is fairly evenly balanced across the curriculum categories, this suggests that Lourdes will generally demonstrate a level of understanding of mathematical concepts at least appropriate for this age, irrespective of whether this is in a format requiring steps in a calculation to be written down, or in mental maths when only 'jottings' or nothing is written. Lourdes is developing the language of mathematics above the expectations for this age. Fluency and agility are better developed than average across both mental maths and applying and understanding maths.
- Where performance is uneven, specific areas of weakness might be addressed as follows:
  - Further targeted practice in the areas identified as being relatively weaker.
  - Practical activities using equipment that is designed to help Lourdes to 'see' the thinking that lies behind any concepts that are not yet secure.
  - Get Lourdes to explain workings to another pupil so that any misconceptions can be highlighted and corrected through discussion.
- Lourdes is generally secure in performing the basic mental calculations expected for this age group. These include age-appropriate fluency with whole numbers, fractions, decimals, percentages and ratio as well as an understanding of the number system, place value (in 4 and 5 digit numbers) and the connections between inverse operations such as multiplication and division.
- Lourdes is developing mathematical reasoning and the ability to solve a wider range of problems, including increasingly complex properties of numbers and arithmetic, and problems demanding efficient written and mental methods of calculation. Lourdes has been introduced to the language of algebra to extend problem solving to a wider variety of contexts and is also developing an understanding of the geometry of increasingly complex shapes and the language needed to describe these.
- Next steps should include opportunities to stretch and develop conceptual understanding, to increase mental agility and to progress the development of formal written mathematics using age-appropriate symbols and methodologies. This can be done by emphasising the higher order skills of hypothesising or predicting (What is an approximate value for 49 × 11?); designing and comparing procedures (How many different ways can we work out 53 – 18? Which is the most efficient?); interpreting results (If 9 × 5 = 45, how many other calculations can we easily work out, e.g. 0.9 × 5?) and applying reasoning (If I have £7.39 in my purse, what coins might they be?).
- In the formal written development there should be an emphasis on developing the correct use of mathematical language, for example in using ':' for ratio and the '=' sign correctly as a mathematical sentence. If algebra has been introduced it is important that Lourdes understands the difference between an 'equation' and an 'expression' and the differences in use of the '=' in each case.
- By providing practice time and frequent opportunities for Lourdes to use one or more known facts to work out more facts, and by engaging Lourdes in discussion and providing opportunities to explain methods and strategies to you and her peers, Lourdes will be helped to make the connections between mental calculations and the written explanations. This will help to develop both problem solving abilities and the language of mathematics, thus improving both mental agility and the ability to plan and carry out accurate calculations in a more formal way.
- More emphasis on explaining methodology, justifying answers and procedures, together with the opportunity to change questions (for example by saying 'What if…' and then altering some aspect of the set question) will ensure that Lourdes develops the problem solving skills of reasoning and generalising.

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Analysis by question

The table below shows each question in the test and the pupil's response (correct/incorrect) compared with the group and standardisation average.

Question number	Curriculum category	Process category	Question content	Score (/x)	Group % correct	Standardisation % correct
MM1	Number	Fluency in facts and procedures	Write in figures the number four hundred and thirty-seven.	1 / 1	79	90
MM2	Number	Fluency in facts and procedures	What number is halfway between three point eight and five point eight?	1 / 1	75	73
MM3	Number	Fluency in conceptual understanding	Write one quarter as a percentage.	1 / 1	42	54
MM4	Number	Mathematical reasoning	Write in figures five hundred and seventy-five correct to the nearest hundred.	1 / 1	88	77
MM5	Number	Fluency in conceptual understanding	Add two point two three and one point zero one.	1 / 1	58	72
MM6	Number	Fluency in conceptual understanding	There are thirty-one days in January. After the seventeenth of January, how many days are left until the end of the month?	1 / 1	50	50
MM7	Measures	Mathematical reasoning	How many weeks are forty-nine days?	1 / 1	29	51
MM8	Number	Fluency in conceptual understanding	What is forty-nine multiplied by a hundred?	1 / 1	54	48
MM9	Number	Fluency in conceptual understanding	I have weeded seventy percent of my path. What percentage is left to weed?	1 / 1	88	89
MM10	Number	Fluency in facts and procedures	Tom has sixteen identical socks. How many pairs of socks does he have?	1 / 1	63	70
MM11	Measures	Fluency in facts and procedures	Bob spends three pounds seventy pence. He pays with a five-pound note. How much change does he get?	1 / 1	46	49
MM12	Number	Fluency in conceptual understanding	The temperature was minus three degrees Celcius. It went up by ten degrees. Write the new temperature.	1 / 1	88	69
MM13	Number	Mathematical reasoning	Forty-eight strawberries are shared between six people. How many does each person get?	1 / 1	54	67
MM14	Number	Fluency in conceptual understanding	One fifth of a number is four. What is the number?	1 / 1	29	52
MM15	Measures	Fluency in facts and procedures	A TV programme starts at ten past five and ends at ten past seven. How many minutes does the programme last?	1 / 1	8	45
AU1	Number	Fluency in conceptual understanding	How far did he run in total on Tuesday and Thursday?	1 / 1	71	67
AU2	Number	Fluency in conceptual understanding	How much further did he run on Saturday than on Wednesday?	1 / 1	21	41

Question number	Curriculum category	Process category	Question content	Score (/x)	Group % correct	Standardisation % correct
AU3	Number	Fluency in conceptual understanding	How far did David run on Wednesday?	0 / 1	42	27
AU4	Statistics	Fluency in conceptual understanding	Which is the youngest cat?	0 / 1	79	83
AU5	Statistics	Mathematical reasoning	What is the difference in mass between the lightest cat and the heaviest cat?	1 / 1	38	46
AU6	Statistics	Problem solving	Write the labels in the correct spaces.	0 / 1	88	82
AU7a	Number	Problem solving	How many packs of books does Miss Begum need to buy?	1 / 1	71	62
AU7b	Number	Fluency in conceptual understanding	How much will she spend?	1 / 1	58	39
AU8	Statistics	Fluency in facts and procedures	How many pencils, pieces of paper and pens does she need for 10 pupils?	1 / 1	75	76
AU9	Statistics	Mathematical reasoning	Write the labels in the spaces on the correct lines.	1 / 1	92	78
AU10	Number	Fluency in conceptual understanding	How much shorter do they need to be?	0 / 1	25	40
AU11	Measures	Mathematical reasoning	Joe buys 4 bananas. What is the smallest amount they can weigh altogether?	1 / 1	54	41
AU12	Measures	Mathematical reasoning	Joe buys 1 banana, 10 grapes and 2 plums. What is the largest amount they can weigh altogether?	1 / 1	29	27
AU13	Measures	Problem solving	Joe buys 500 grams of grapes. What is the largest number of grapes he can get?	1 / 1	21	23
AU14a	Number	Fluency in conceptual understanding	Circle the name of the city that is warmest in winter.	1 / 1	88	77
AU14b	Number	Mathematical reasoning	Cardiff is 8 degrees warmer than Oslo. What is the temperature in Cardiff?	1 / 1	79	61
AU15	Measures	Fluency in conceptual understanding	How many cubes does he use? Use the cards to show how to work it out. Write your answer	1 / 2	17	33
AU16a	Number	Fluency in conceptual understanding	Which is the odd one out?	1 / 1	33	59
AU16b	Number	Mathematical reasoning	Write another fraction with the same value as the other 4 fractions.	1 / 1	25	38
AU17	Number	Problem solving	How many children are there at the camp?	2 / 2	15	22
AU18a	Number	Mathematical reasoning	Which rod is two-thirds of rod B?	1 / 1	21	38
AU18b	Number	Mathematical reasoning	Which rod is 50 per cent of rod E?	1 / 1	46	57
AU18c	Number	Mathematical reasoning	Fill in the fraction.	0 / 1	17	25
AU19a	Number	Fluency in conceptual understanding	Write these fractions in the correct boxes.	2 / 2	52	61
AU19b	Number	Mathematical reasoning	Make up one more fraction for each box.	1 / 1	42	55
AU20	Number	Problem solving	How many adults are needed on the trip?	1 / 1	50	48
AU21	Geometry	Mathematical reasoning	Circle all the squares that are cut into 2 equal pieces.	1 / 2	65	56
AU22	Geometry	Mathematical reasoning	Circle all the squares where the cut is a line of symmetry.	1 / 1	25	40

Question number	Curriculum category	Process category	Question content	Score (/x)	Group % correct	Standardisation % correct
AU23	Number	Fluency in conceptual understanding	Work out the other ways and fill in the gaps below.	1 / 2	13	18
AU24	Number	Mathematical reasoning	How many boxes do they need to buy? How many bags of crisps will be left over?	1 / 1	58	44
AU25	Number	Mathematical reasoning	Put the correct numbers in the spaces.	1 / 3	50	57
AU26	Measures	Fluency in conceptual understanding	What is the perimeter of her vegetable garden?	1 / 1	38	36
AU27a	Measures	Mathematical reasoning	Which diagram shows a vegetable garden that has the same area, but a smaller perimeter?	1 / 1	21	33
AU27b	Measures	Fluency in conceptual understanding	What is the perimeter of the new vegetable garden?	1 / 1	13	29
AU28a	Number	Fluency in conceptual understanding	Sort the cities in order of distance from London.	0 / 2	69	71
AU28b	Number	Mathematical reasoning	How much further away from London is Honolulu than Miami?	1 / 1	21	39
AU29	Geometry	Fluency in facts and procedures	Which is the smallest angle?	1 / 1	75	66
AU30	Geometry	Fluency in facts and procedures	Which is the largest angle?	0 / 1	58	52
AU31	Geometry	Fluency in facts and procedures	Which 2 angles are reflex angles?	1 / 1	46	46
AU32	Geometry	Fluency in conceptual understanding	Which 2 angles together would make 1 complete turn?	1 / 1	25	41
AU33i	Statistics	Mathematical reasoning	Draw the correct scores on the bar chart.	1 / 1	63	71
AU33ii	Statistics	Mathematical reasoning	Draw the correct scores on the bar chart.	1 / 1	50	53
AU33iii	Statistics	Mathematical reasoning	Draw the correct scores on the bar chart.	1 / 1	46	38